

Overview

- 1) Two more examples
counting & properties of probability
- 2) Random Variables

of ways to
Partition set of distinct
objects into r distinct
subsets of sizes
 $n_1, n_2, \dots, n_r$ is
$$\frac{n!}{n_1! n_2! \dots n_r!}$$

A football team consists of 20 defensive & 20 offensive players. The players are supposed to be paired in groups of 2 for allocation as roommates. If the pairing is done at random, what is the prob that there are n_o offensive-defensive pairs

$$\text{Total number of ways to do the pairing} = \frac{40!}{(2!)^{20} 20!}$$

{1,2} {3,4} ... {39,40}
" " " " " "
{3,4} {1,2} ... {39,40}

of ways offensive
defensive players
are paired among
themselves $\Rightarrow$
no cross pairing

$$= \left[\frac{20!}{(2!)^{10} 10!} \right] \left[\frac{20!}{(2!)^{10} 10!} \right]$$

of ways in which
each defensive player
is paired with
offensive
player

$$= 20!$$

Probability that there
are no offensive defensive pairs



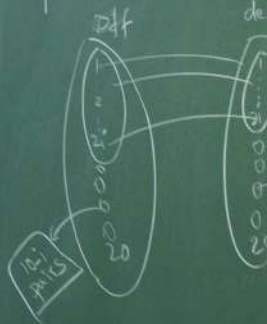
$$= \left(\frac{20!}{(2!)^{10} 10!} \right)^2 \bigg/ \left(\frac{40!}{(2!)^{20} 20!} \right)$$

Probability that every pair is
an offensive-defensive

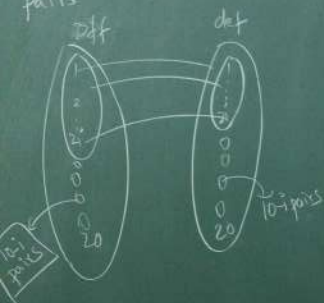
$$\text{pair} = \frac{(20!)}{\left(\frac{40!}{(2!)^{20} 20!} \right)}$$

Probability that there are
exactly 2 offensive-defensive
roommate pairs ^{for some} $\in \{1, 2, \dots, 10\}$

of ways to match st
there are exactly
2 offensive-defensive
pairs



of ways to match st
there are exactly
 $2i$ offensive-defensive
pairs



ways to
choose $2i$ elements
each from off, def set

$$= \binom{20}{2i} \binom{20}{2i} (2i)! \left[\frac{(20-2i)!}{(2i)^{10-i} (10-i)!} \right]^2$$

of ways to pair the chosen
 $2i$ members

If $i=10$ $= 20!$ (check)

$i=0$ $= \left[\frac{20!}{(2i)^{10} 10!} \right]^2$

Class

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B|A)$$

additivity

$$\begin{aligned} P(B) &= P(B|A) + P(B|A^c) \\ &= P(B|A) + P(B \cap A^c) \end{aligned}$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(B \cap A)$$

$$P(A^c) = 1 - P(A)$$

Additivity $P(A \cup A^c) = P(A) + P(A^c)$

Nonnegativity $P(A) \geq 0$

Normalization $P(\Omega) = 1$



Let $A_1, A_2, \dots, A_n$ be some events

$$P(A_1 \cup A_2 \cup \dots \cup A_n) \quad \text{[Inclusion \& Exclusion Principle]}$$

$$= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k)$$

$$- \dots - (-1)^{j-1} \sum_{1 \leq i_1 < i_2 < \dots < i_j \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_j}) + \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

Proof: True for $n=2$. Assume it is true for $n=k-1$.

We'll show that it holds for $n=k$ case

$$\begin{aligned} P(\underbrace{A_1 \cup A_2 \cup \dots \cup A_{k-1}}_{=B} \cup A_k) &= P(B \cup A_k) \\ &= P(B) + P(A_k) - P(B \cap A_k) \rightarrow (1) \end{aligned}$$

$$\begin{aligned} P(B \cap A_k) &= P\left(\left(\bigcup_{i=1}^{k-1} A_i\right) \cap A_k\right) \\ &= P\left(\bigcup_{i=1}^{k-1} (A_i \cap A_k)\right) \end{aligned}$$

From induction assumption

$$\begin{aligned} P\left(\bigcup_{i=1}^{k-1} (A_i \cap A_k)\right) &= \sum_{i=1}^{k-1} P(A_i \cap A_k) - \sum_{1 \leq i_1 < i_2 \leq k-1} P(A_{i_1} \cap A_{i_2} \cap A_k) \rightarrow (2) \\ &\quad + \dots + (-1)^{k-2} P(A_1 \cap A_2 \cap \dots \cap A_{k-1} \cap A_k) \end{aligned}$$



From ① & ②

$$\begin{aligned}
 & P(A_1 \cup A_2 \cup \dots \cup A_k) \\
 &= P(A_k) + \left(\sum_{i=1}^{k-1} P(A_i) - \sum_{1 \leq i_1 < i_2 \leq k-1} P(A_{i_1} \cap A_{i_2}) \right. \\
 &\quad \dots \left. (-1)^{k-2} P(A_1 \cap A_2 \cap \dots \cap A_{k-1}) \right. \\
 &\quad \left. - \sum_{1 \leq i \leq k-1} P(A_i \cap A_k) + \sum_{1 \leq i_1 < i_2 \leq k-1} P(A_{i_1} \cap A_{i_2} \cap A_k) \right. \\
 &\quad \left. \dots (-1)^{k-1} P(A_1 \cap A_2 \cap \dots \cap A_k) \right)
 \end{aligned}$$

→ ①

$(A_1 \cap A_2 \cap \dots \cap A_k) \rightarrow ②$

$P(A_1 \cap A_2 \cap \dots \cap A_k)$

$$\begin{aligned}
 &= \sum_{i=1}^k P(A_i) - \\
 &\quad + \sum_{1 \leq i_1 < i_2 \leq k} P(A_{i_1} \cap A_{i_2}) - \dots
 \end{aligned}$$

$$= \sum_{i=1}^k P(A_i) - \sum_{1 \leq i_1 < i_2 \leq k} P(A_{i_1} \cap A_{i_2})$$

$$+ \sum_{1 \leq i_1 < i_2 < i_3 \leq k} P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) - \dots (-1)^{k+1} P(A_1 \cap A_2 \cap \dots \cap A_k)$$

← x →

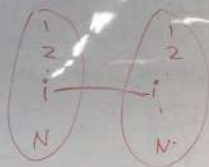
Suppose each of the N people at a party throw their hat into the center of the room. Each person randomly selects a hat. What is the probability that none of them select their own hat?

E_i : i th person is paired with their own hat.

$$P(E_1^c \cap E_2^c \cap \dots \cap E_N^c)$$

$$= 1 - P(E_1 \cup E_2 \cup \dots \cup E_N)$$

$$P(E_i) = \frac{(N-1)!}{N!} = \frac{1}{N}, \quad \forall i=1, 2, \dots, N$$



Outcomes can be thought of as bijective maps.

$N!$ possibilities.

$$P(E_i \cap E_j) = \frac{(N-2)!}{N!} = \frac{1}{N(N-1)}, \quad P(E_1 \cap E_2 \cap \dots \cap E_r) = \frac{(N-r)!}{N!}$$

$$\begin{aligned} P(E_1 \cup E_2 \cup \dots \cup E_N) &= \sum_{i=1}^N P(E_i) - \sum_{1 \leq i < j \leq N} P(E_i \cap E_j) \\ &\quad + \dots + (-1)^{N-1} P(E_1 \cap E_2 \cap \dots \cap E_N) \\ &= N \times \frac{1}{N} - \binom{N}{2} \frac{(N-2)!}{N!} + \dots + (-1)^{N-1} \binom{N}{N} \frac{(N-N)!}{N!} \\ &\quad \dots + (-1)^{N-1} \frac{1}{N!} \\ &= 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots + (-1)^{N-1} \frac{1}{N!} + \dots + (-1)^{N-1} \frac{1}{N!} \end{aligned}$$

$$P(E_1^c \cap E_2^c \cap \dots \cap E_N^c) = 1 - \left(1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots + (-1)^{N-1} \frac{1}{N!}\right) = \frac{1}{N!}$$

Alternate approach (Suggested by students in class).

Count the # of ways
matching can be done
s.t none of them gets
matched to their hats.

(Ref: Sheldon M. Ross
A first course in
probability).
Problem 17
chapter 2.

$A_N :=$ # of ways matching can be done for N people w.r.t N hats st none of them get matched to their corresponding hats.

Number of ways first person can select a hat without a match $= N-1$

Suppose person 1 selected hat i .

of ways remaining people $\{2, \dots, N\}$ select their hats such that ~~no~~ one gets a match

$A_{N-2} = \begin{cases} i\text{-th person gets hat } 1 \\ \text{and rest don't have a match} \end{cases}$

$A_{N-1} = \begin{cases} i\text{-th person doesn't get hat } 1 \\ \text{and rest don't have a match} \end{cases}$

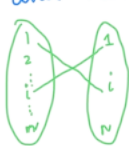
We are looking for a matching from $\{2, 3, \dots, N\}$

to $\{1, 2, \dots, i-1, i+1, \dots, N\}$ st i is not mapped to 1 and

2 not mapped to 2
3 not " to 3
 $\vdots$
 N not " to N .

of such mappings $= A_{N-1}$.

$$\Rightarrow A_N = (A_{N-1} + A_{N-2})(N-1).$$



Find the probability that exactly k of them get their own hats

$$\begin{aligned}
 & P(E_1 \cap E_2 \cap \dots \cap E_k \cap E_{k+1}^c \cap \dots \cap E_N^c) \\
 &= P(E_1 \cap \dots \cap E_k) P(B | E_1 \cap \dots \cap E_k) \\
 &= \frac{(N-k)!}{N!} P_{N-k}
 \end{aligned}$$

$$\sum_{\substack{S \subseteq [N] \\ |S|=k}} P\left(\bigcap_{i \in S} E_i\right) P\left(\bigcap_{i \notin S} E_i^c\right) = \binom{N}{k} \frac{(N-k)!}{N!} P_{N-k} = \frac{P_{N-k}}{k!}$$



Random Variables (RV)

RVs are defined as a function from sample space (Ω)
to set of real numbers ($\mathbb{R}$)
 $X: \Omega \rightarrow \mathbb{R}$

(a) # of heads in 3 coin tosses.

$$\Omega = \{HHH, HHT, THH, \dots\}$$

$$X(\omega) = \begin{cases} 3 \\ 0 \\ 1 \\ 2 \end{cases}$$

$$\omega \in \{HHH\}$$

$$\omega \in \{TTT\}$$

$$\omega \in \{HHT, THT, TTH\}$$

$$\omega \in \{THH, HHT, HTH\}$$

(b) Given $\Omega = [0, 1]$

$$P(E_1 | E_2 \cap E_3) = \frac{(N-1)!}{n!}$$

$$X(\omega) = \begin{cases} 0 & \omega \leq 0.5 \\ 1 & \omega > 0.5 \end{cases}$$

(c) $X(\omega) = \omega \quad \omega \in [0, 1] = \Omega$

The values that the random variable takes (ie, the range of RV function) can be countable or uncountable.